

STATEMENT OF WORK

1. Objective/Requirements

The 10x10 SWT requires the measurement of tunnel dewpoint temperature to accurately correct test section operating conditions for the presence of water vapor. Measurement of this value is critical to data quality in the facility. The Stephens Analytical MicroView Mini 6000 hygrometers provide a combination of ease-of-use, accuracy, and response time not found elsewhere through searches among common instrumentation makers/vendors.

2. Characteristics, Scope, and Specs

The contractor that provides this instrumentation shall perform the following:

- Supply four hygrometers with the following capabilities/characteristics:
 - Range of 5-6000 ppm(V)
 - Temperature-controlled silicon chip moisture sensor
 - Automated Push Purge system to enable fast response time
 - 4-20 mA linear output
 - NIST traceable calibration.

3. Place of Performance

10x10 Supersonic Wind Tunnel at the NASA Glenn Research Center in Cleveland, OH.

4. Period of Performance

Testing may begin with this hygrometer 2026 and will be used for the lifetime of the unit as the primary dewpoint temperature measurement device(s) for the facility.